IIIT Sri City

Course: IDHV 2026M

College Student Digital Lifestyle Profile

A survey-based exploratory data analysis of screen time, study habits and academic performance

Project Report

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Sample: 83 responses collected; 78 analysed after cleaning

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1. Introduction

1.1 Research Question

College students spend a lot of time on digital activities — social media, gaming, video, online shopping, and now AI tools — alongside their studies. This report asks one main question:

Research question: How do students' digital habits relate to their grades, study habits, focus and productivity?

To answer it, we looked at five smaller questions:

- Does more screen time, social media or gaming go with a lower CGPA or less study time?
- Does checking the phone during study sessions affect focus and productivity?
- Do sleep length and a regular sleep schedule affect focus and productivity?
- How often, and for what, do students use AI tools — and does this affect CGPA?
- How happy are students with their balance between screens, studies and personal time?

1.2 Motivation

Digital devices are now part of everyday student life. Many people assume heavy screen time hurts grades, but this is rarely tested directly, and the real picture can be more mixed — for example, AI tools may help studying rather than hurt it. Earlier studies found that multitasking on media while studying lowers learning (Rosen et al., 2013; Junco, 2012), that simply having a phone nearby reduces available thinking capacity (Ward et al., 2017), that screen use is linked to shorter sleep (Hale & Guan, 2015), and that an irregular sleep schedule weakens academic performance (Phillips et al., 2017). These ideas are rarely tested together in one group of students. This report tests them together, using real survey data instead of guesswork.

2. Methodology

2.1 Research Design

This is a one-time (cross-sectional) survey study, not an experiment. We collected answers once from each student and did not control any conditions. So we can find **links** between habits and outcomes, but we cannot prove that one causes the other.

2.2 Data Collection

We used an online form with 18 main questions covering: age and year of study; CGPA; daily hours on screens, social media, gaming and OTT/YouTube; AI tool use and purpose; online shopping; phone checking during study; study hours; sleep hours and sleep regularity; and five 1–5 ratings for productivity, focus and satisfaction with balance. The questions mix numbers (CGPA, hours, counts) with categories and ratings (year of study, main free-time activity, AI purposes).

The form also exported four metadata columns — timestamp, participant ID, participant type and completion time. All four were dropped before analysis. The raw file (**raw_data.csv**) holds **83 responses**, above our target of 50. The full wording of every question is in Appendix A.

Table 1 lists every column of **raw_data.csv** — the 4 metadata columns and the 18 questions — with its meaning. Long question headings are shortened with …; the full wording and answer options are in Appendix A. The columns created during preprocessing are described in Chapter 3.

Table 1. Columns of the raw data and their meaning.

Column in raw_data.csv	Meaning
Timestamp (UTC)	Date and time the form was submitted
participant_id	Anonymous ID given to each respondent
participant_type	Type of respondent (e.g. registered)
duration_seconds	Seconds taken to fill in the form
What is your age?	Age group of the student
What is your current year of study?	Year of study (2nd, 3rd or 4th Year)
What is your current CGPA? …	CGPA on a 10-point scale
… total screen time do you have?	Total screen hours on a typical day
… hours do you spend on social media …	Social media hours per day
… hours do you spend gaming …	Gaming hours per day
… watching YouTube/OTT …	YouTube / OTT hours per day
How often do you use AI tools …	How often AI tools are used
What do you mainly use AI tools for?	Purposes of AI use (multi-select text)
How many times do you shop online …	Online purchases per month
… check your phone during a normal study session?	Phone checking while studying (1–5)
… hours do you spend studying outside of classes?	Self-study hours per day
How many hours do you usually sleep per night?	Sleep hours per night
… describe your usual sleep schedule?	Regularity of sleep schedule (1–5)
How productive do you usually feel …	Self-rated productivity (1–5)
How focused are you usually while studying?	Self-rated focus (1–5)
How satisfied are you … balance …	Satisfaction with digital–study–personal balance (1–5)
Which activity takes up most of your free time?	Main free-time activity (7 categories)

2.3 Sampling Method

We used convenience sampling: the form was shared in student group chats and with friends, and anyone could choose to respond. We picked this because we had no full list of students to sample randomly from, this is an early exploratory study rather than one meant to represent all students, and we had no time or budget to chase a random sample. The trade-off: our sample likely over-represents students who are comfortable with technology, so results describe this group only (see Section 5.2).

2.4 Analysis Plan

Most of our variables are counts, hours, or 1–5 ratings, not smooth bell-curve data. So we mainly used rank-based statistics, which don't assume a normal distribution:

- **Spearman's correlation (ρ)** to measure how two variables move together; we call a result significant when $p < 0.05$.
- **D'Agostino–Pearson test** to check whether each numeric variable is normally distributed, which supports the choice of rank-based methods.
- **Descriptive statistics and charts** (histograms, bar charts, box plots, scatter plots and a correlation matrix) to explore the data.

All steps were done in Python (pandas, NumPy, SciPy, Matplotlib, Seaborn; McKinney, 2010; Virtanen et al., 2020; Hunter, 2007; Waskom, 2021) in one notebook, **AnalysisCode.ipynb**. Every table and figure in this report comes from that notebook or from **cleaned_data.csv**, the file it saves.

3. Data Preprocessing

3.1 Data Organization

We dropped the four metadata columns and renamed the remaining 18 columns with short, clear labels such as **screen_time_hrs** and **focus_during_study** (Table 1 lists the original columns). The raw data had 83 rows, **no missing values** and **no duplicate responses**, so nothing had to be filled in or merged.

3.2 Removing Invalid Responses

We removed responses in four steps. Each one is checked in the notebook before it is applied, so every removal can be traced:

Table 2. Responses removed during cleaning.

Step	What was removed	Rows removed	Rows left
1	Impossible combination: age "16 or below" with "4th Year" of study	1	82
2	Invalid CGPA: 1.7 on a 10-point scale (we kept only CGPA above 6.0)	2	80
3	Contradictory AI answer: "I don't use AI tools" chosen together with specific AI uses	1	79
4	Extreme response: 16 h social media, 16 h gaming, 13 h OTT, 16 h study, 16 h sleep and 26 online purchases a month, all from one student	1	78

83 responses collected; 5 removed in four steps; 78 kept for analysis (94%)

3.3 Checking the Screen-Time Questions

Social media, gaming and OTT time should each be part of total screen time, so together they should not add up to more than the reported total. Among the 78 kept responses, this is broken in **30 of them (38.5%)**.

We kept these responses and only flagged them. The activities can overlap with other screen use (multitasking), and the question never said whether the parts count toward the total, so a student may have read "total screen time" as time *beyond* social media, gaming and OTT. Removing 38.5% of a small sample would do more harm than good. The screen-time correlations in Section 4.2.4 should be read with this in mind.

3.4 Encoding and Reverse-Coding

We turned the answers into numbers the analysis can use:

- **Year of study** became an ordered category (2nd < 3rd < 4th). **Age** stays as the original answer categories (17 to "23 or above").
- **AI-use frequency** became 0–3 (Never, Rarely, Daily, Several times a day).
- **AI purposes** (pick many) became nine yes/no columns; **main free-time activity** became seven.

Three questions were scored so that 1 meant "best": focus, productivity and sleep regularity (for example, 1 = very focused, 5 = not focused). That is easy to misread, so we flipped them (6 − score) into **focus_score**, **productivity_score** and **sleep_regularity_score**; a check in the notebook confirms each new column is the exact mirror of the original ($\rho = -1$). **Throughout this report, a higher score always means better.**

Table 3. How each rating question is coded, and what the notebook does with it.

Original column	Original coding	Analysis column	New coding
focus_during_study	1 = Very focused, 5 = Not focused	focus_score	1 = Not focused, 5 = Very focused
productive_in_study	1 = Very productive, 5 = Not productive	productivity_score	1 = Not productive, 5 = Very productive
sleep_schedule	1 = Very regular, 5 = Very irregular	sleep_regularity_score	1 = Very irregular, 5 = Very regular
phone_during_study	1 = Not at all, 5 = Extremely	unchanged	Higher = more phone use (more distraction)
satisfication_of_digital_profile	1 = Very dissatisfied, 5 = Very satisfied	unchanged	Higher = more satisfied

Two rating questions already read "higher = more" and were **left unchanged**: **phone_during_study** (1 = Not at all, 5 = Extremely, so a higher number means more phone use and more distraction) and **satisfication_of_digital_profile**. The original coded columns are kept alongside the three new ones.

3.5 Outlier Detection and Treatment

We used the IQR method (values beyond Q1 − 1.5×IQR or Q3 + 1.5×IQR) to flag unusual values in the hours and count columns. It was run on the 79 rows left after step 3 of Table 2, before the extreme response (step 4) was removed:

Table 4. IQR outlier check on the hours and count columns (79 rows). Output of the notebook's outlier_summary.

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	Outlier Count
screen_time_hrs	5.0	10.0	5.0	-2.5	17.5	0
social_media_time_hrs	2.0	5.0	3.0	-2.5	9.5	4
gaming_time_hrs	0.0	1.0	1.0	-1.5	2.5	10
ott_time_hrs	1.0	3.0	2.0	-2.0	6.0	5
monthly_online_shopping	1.0	3.0	2.0	-2.0	6.0	6
phone_during_study	3.0	4.0	1.0	1.5	5.5	1
study_time	2.0	4.0	2.0	-1.0	7.0	4
sleep_time_hrs	6.0	8.0	2.0	3.0	11.0	1

Gaming has many flagged values only because most students answered 0 or 1, so these are not errors. Instead of deleting rows — which would waste data, and the values are unusual but not impossible — we **capped** them at sensible limits (winsorization) on the final 78 rows: at the IQR fences for five columns, and at the 95th percentile for gaming and online shopping, where the IQR is almost zero. The phone-use column is a 1–5 rating and was not capped.

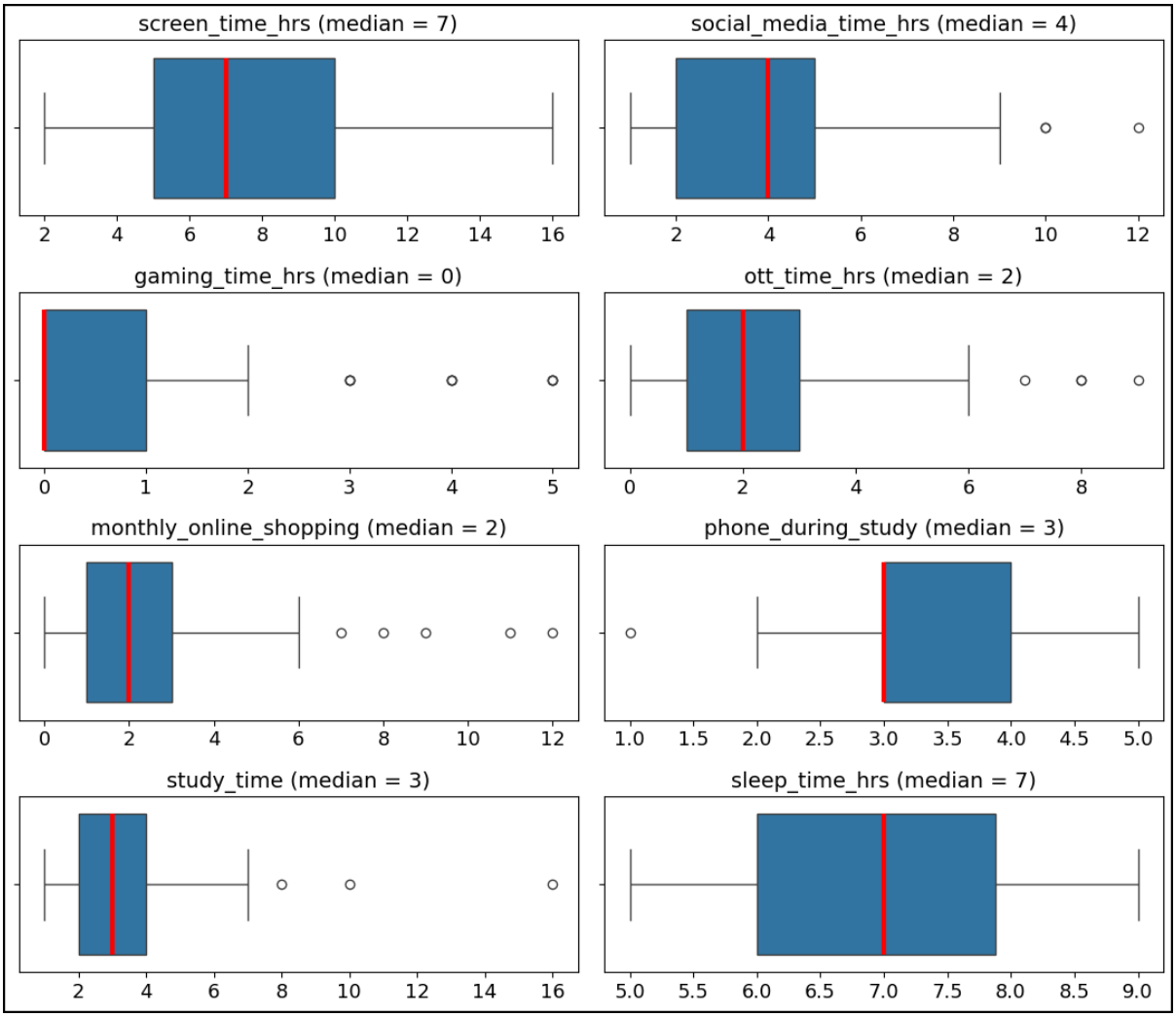


Figure 1. *Box plots of the eight hours, count and rating columns on the 78 kept rows, before capping.*

Table 5. Caps applied to each column, and how many values were changed (notebook output).

Column	Lower cap	Upper cap	Values capped	Max before	Max after
screen_time_hrs	2.0	17.50	0	16.0	16.00
social_media_time_hrs	1.0	9.50	3	12.0	9.50
ott_time_hrs	0.0	6.00	4	9.0	6.00
study_time	1.0	7.00	3	16.0	7.00
sleep_time_hrs	5.0	10.69	0	9.0	9.00
gaming_time_hrs	0.0	4.00	3	5.0	4.00
monthly_online_shopping	0.0	7.15	4	12.0	7.15

Capping does not change the *order* of values, so the rank-based (Spearman) results barely move. It mainly affects the averages and spreads in Table 8. All results in Chapter 4 use the capped data.

3.6 Normalization and Scaling

Our variables use very different scales (CGPA out of 10, hours up to 16, 1–5 ratings). So we also made **min–max (0–1)** and **z-score** versions of the eight numeric variables (CGPA, screen, social media, gaming, OTT, shopping, study time, sleep). We also made a **log** version of the three variables with skew above 1: gaming (1.46), OTT (1.10) and online shopping (1.08). The scaled table (78 rows × 19 columns) stays inside the

notebook; every table and chart in this report uses the original, easy-to-read units. The cleaned and capped data (78 rows × 38 columns) is saved as **cleaned_data.csv**.

Table 6. Skewness of the eight numeric variables before and after a log1p transform (notebook output). A log version is kept only where the original skew is above 1.

Variable	skew	skew_after_log1p
cgpa	-0.19	-0.36
screen_time_hrs	0.45	-0.36
social_media_time_hrs	0.81	-0.13
gaming_time_hrs	1.46	0.73
ott_time_hrs	1.10	0.01
monthly_online_shopping	1.08	-0.03
study_time	0.62	-0.03
sleep_time_hrs	0.23	-0.00

The log transform pulls the three lopsided variables closer to symmetric: gaming from 1.46 to 0.73, OTT from 1.10 to 0.01 and online shopping from 1.08 to −0.03. The other five variables are not strongly skewed, so they keep their original scale. In the scaled table, the min–max columns run from 0 to 1 and the z-score columns have a mean of 0 and a standard deviation of about 1 (1.01 in the notebook summary).

Table 7. Summary of the 19 columns in df_scaled (78 rows × 19 columns; notebook output).

Column	mean	std	min	max
cgpa_minmax	0.50	0.22	0.00	1.00
cgpa_z	0.00	1.01	-2.23	2.26
screen_time_hrs_minmax	0.37	0.23	0.00	1.00
screen_time_hrs_z	-0.00	1.01	-1.63	2.76
social_media_time_hrs_minmax	0.34	0.27	0.00	1.00
social_media_time_hrs_z	0.00	1.01	-1.30	2.49
gaming_time_hrs_minmax	0.22	0.30	0.00	1.00
gaming_time_hrs_z	0.00	1.01	-0.74	2.60
ott_time_hrs_minmax	0.35	0.28	0.00	1.00
ott_time_hrs_z	-0.00	1.01	-1.25	2.33
monthly_online_shopping_minmax	0.29	0.28	0.00	1.00
monthly_online_shopping_z	-0.00	1.01	-1.07	2.57
study_time_minmax	0.35	0.29	0.00	1.00
study_time_z	0.00	1.01	-1.22	2.21
sleep_time_hrs_minmax	0.49	0.25	0.00	1.00
sleep_time_hrs_z	-0.00	1.01	-2.01	2.08
gaming_time_hrs_log1p	0.47	0.55	0.00	1.61
ott_time_hrs_log1p	1.00	0.53	0.00	1.95
monthly_online_shopping_log1p	0.93	0.65	0.00	2.10

4. Results and Analysis

78 students analysed; 0.49 focus–productivity link (ρ); −0.37 phone use vs. focus (ρ); 2.41 satisfaction with balance (of 5)

4.1 Descriptive Statistics

Table 8. Descriptive statistics for every analysed variable (n = 78, after capping; notebook output).

Variable	mean	median	mode	std	min	max	IQR	CV %	skew	kurtosis
cgpa	8.29	8.3	7.7	0.76	6.6	10.00	1.20	9.21	-0.19	-0.50
screen_time_hrs	7.21	7.0	6.0	3.21	2.0	16.00	5.00	44.59	0.45	-0.33
social_media_time_hrs	3.92	4.0	3.0	2.25	1.0	9.50	3.00	57.55	0.81	0.15
gaming_time_hrs	0.88	0.0	0.0	1.21	0.0	4.00	1.00	136.31	1.46	1.27
ott_time_hrs	2.10	2.0	1.0	1.69	0.0	6.00	2.00	80.24	1.10	0.41
monthly_online_shopping	2.10	2.0	1.0	1.98	0.0	7.15	2.00	94.48	1.08	0.62
study_time	3.13	3.0	2.0	1.76	1.0	7.00	2.00	56.29	0.62	-0.47
sleep_time_hrs	6.97	7.0	6.0	0.98	5.0	9.00	1.88	14.13	0.23	-0.70
sleep_regularity_score	2.69	3.0	3.0	1.07	1.0	5.00	1.00	39.85	0.20	-0.50
productivity_score	2.97	3.0	3.0	0.98	1.0	5.00	1.00	32.95	0.14	0.03
focus_score	2.95	3.0	3.0	0.97	1.0	5.00	2.00	32.75	0.10	-0.40
satisfication_of_digital_profile	2.41	2.0	3.0	0.93	1.0	5.00	1.00	38.66	0.17	-0.34
phone_during_study	3.38	3.0	3.0	0.96	1.0	5.00	1.00	28.26	-0.03	-0.56

What the numbers tell us:

- **CGPA is steady** across students: mean 8.29, and it varies little (CV about 9%).
- **Screens beat study.** Screen time averages 7.2 hours a day, more than double the 3.1 hours of study. Social media (3.9 h) is the largest single category, then OTT (2.1 h) and gaming (0.9 h).
- **Gaming, shopping and OTT are lopsided:** most students do little, a few do a lot (CV 80–136%). For these, the median (0–2) describes a typical student better than the mean.
- **Sleep is tightly clustered** (mean 7.0 h, CV about 14%).
- **Balance is the weak spot.** All 1–5 ratings sit near the middle, but satisfaction with digital/study/life balance is the lowest at 2.41 out of 5.

Table 9. D'Agostino–Pearson normality test for the eight numeric variables (H0: the data are normally distributed; notebook output).

Variable	Statistic	p-value	Normal at 5%?
cgpa	1.5905	0.4515	Yes
screen_time_hrs	3.0675	0.2157	Yes
social_media_time_hrs	8.1704	0.0168	No
gaming_time_hrs	23.0469	0.0000	No
ott_time_hrs	13.6694	0.0011	No
monthly_online_shopping	13.9289	0.0009	No
study_time	5.8509	0.0536	Yes
sleep_time_hrs	3.7896	0.1503	Yes

CGPA, screen time, study time and sleep are not rejected as normal ($p > 0.05$); social media, gaming, OTT and shopping clearly are ($p < 0.05$), matching their skew. Because most variables are skewed counts or 1–5 ratings, we use rank-based Spearman correlation throughout.

4.2 Exploratory Data Analysis

4.2.1 How the Key Variables Are Spread

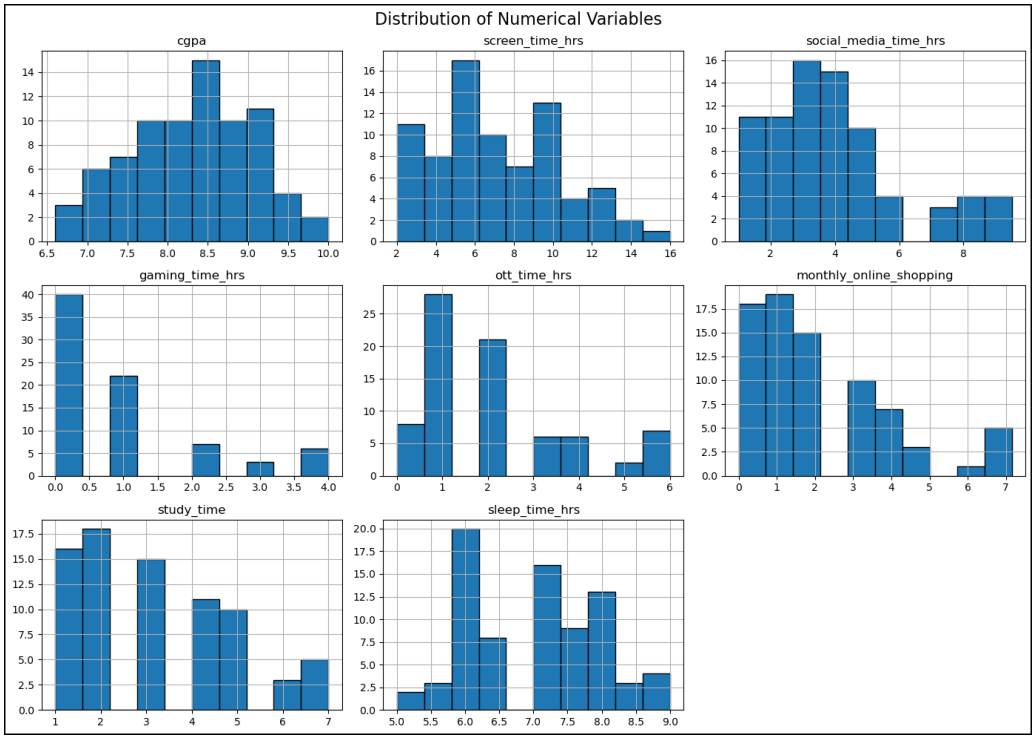


Figure 2. *Distributions of the eight numeric variables after cleaning and capping.*

Most students game very little, while CGPA and sleep look closer to a bell curve. Screen time has two peaks — one around 5–6 hours and a smaller one around 9–10.

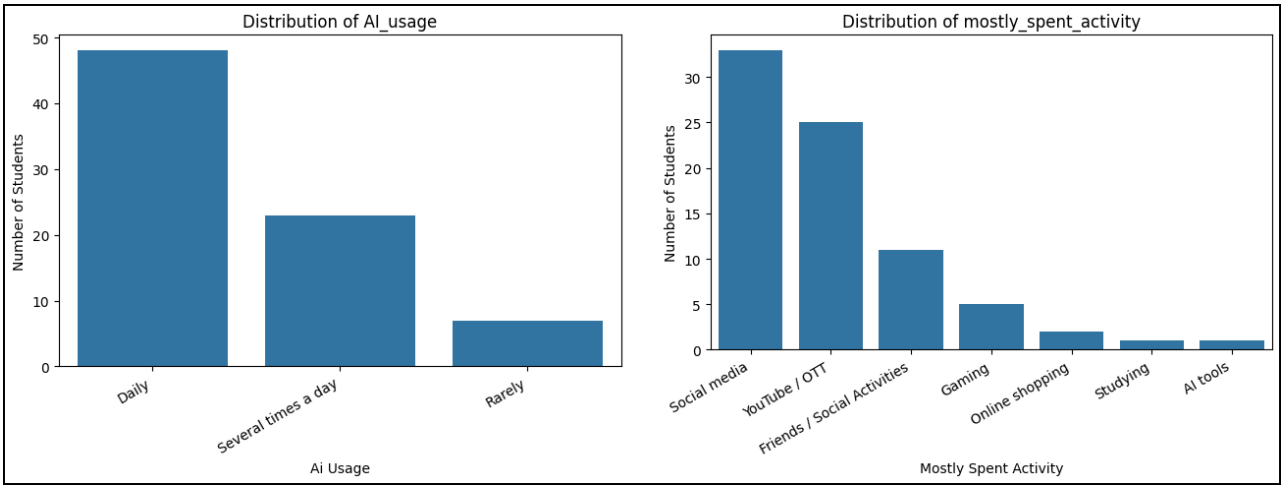


Figure 3. *AI tool usage frequency (left) and main free-time activity (right).*

Of the 78 students, 46 (59.0%) are in 3rd year, 21 (26.9%) in 2nd and 11 (14.1%) in 4th. Almost everyone uses AI tools: 48 students (61.5%) daily and 23 (29.5%) several times a day; only 7 (9.0%) use them rarely. Social media (33 students, 42.3%) and YouTube/OTT (25 students, 32.1%) are the top two free-time activities, ahead of friends and social activities (14.1%) and gaming (6.4%).

4.2.2 Digital Habits and CGPA

None of the six habits we tested is significantly linked to CGPA (all p > 0.05). Study time is the strongest (ρ = 0.140) and even that is weak.

Table 10. Spearman correlation of each habit with CGPA (notebook output).

Variable	Spearman Correlation	P-value
study_time	0.140	0.2215
gaming_time_hrs	-0.010	0.9317
screen_time_hrs	-0.012	0.9141
ott_time_hrs	-0.098	0.3935
social_media_time_hrs	-0.113	0.3264
sleep_time_hrs	-0.140	0.2206

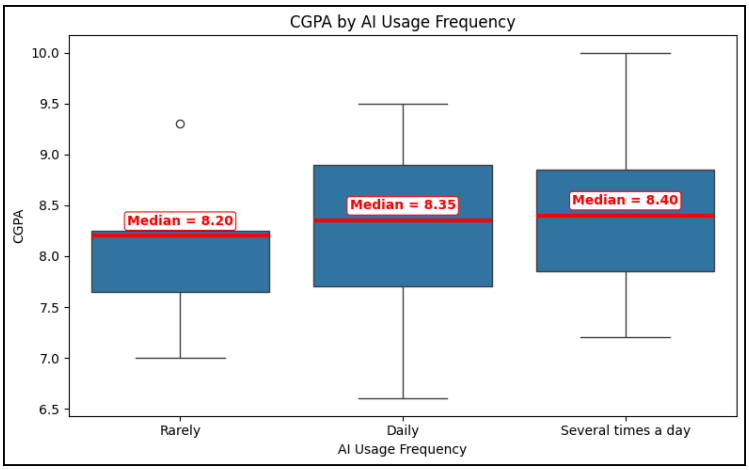


Figure 4. *CGPA by AI tool usage frequency.*

CGPA also looks similar across AI-use groups: the median is 8.20 for rare users, 8.35 for daily users and 8.40 for several-times-a-day users. The number of different AI purposes a student uses is unrelated to CGPA (ρ = 0.005, p = 0.9649), and AI-use frequency shows only a weak link in the correlation matrix (ρ = 0.15, Figure 6).

4.2.3 Focus, Productivity and Phone Use

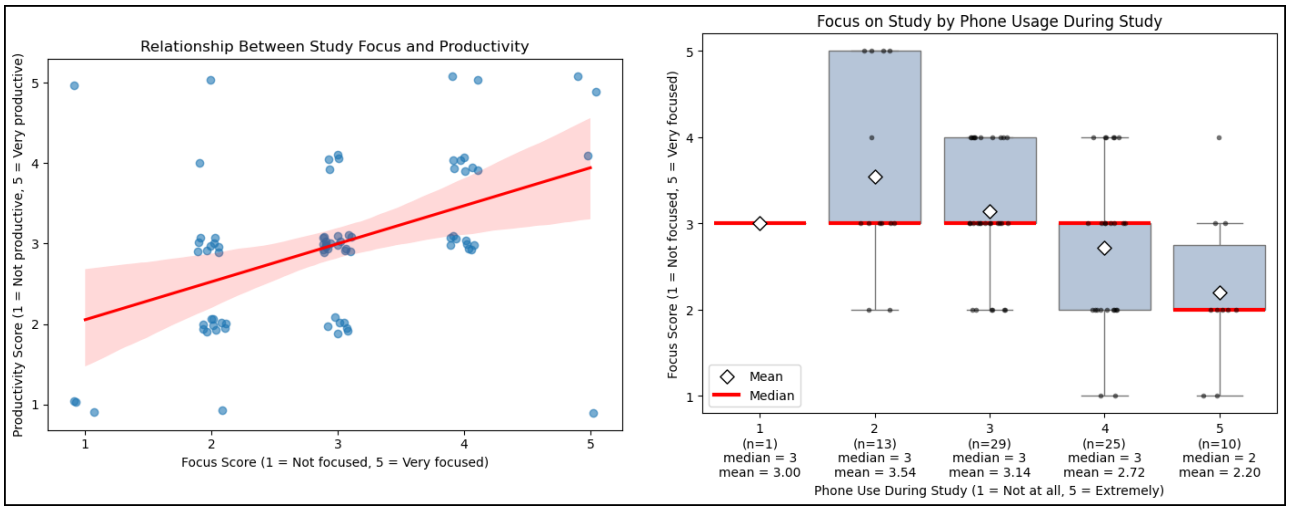


Figure 5. *Focus against productivity (left) and focus by phone use during study (right). Both scores run from 1 (worst) to 5 (best). Dots are individual students, diamonds are group means and red lines are medians. For phone-use levels 1 to 5 the median focus score is 3, 3, 3, 3 and 2 and the mean is 3.00, 3.54, 3.14, 2.72 and 2.20 (n = 1, 13, 29, 25 and 10).*

Table 11. Spearman correlations with focus and productivity (notebook output).

Variable	Outcome	Spearman Correlation	P-value
focus_score	productivity_score	0.486	0.0000
productivity_score	focus_score	0.486	0.0000
phone_during_study	focus_score	-0.374	0.0007
phone_during_study	productivity_score	-0.237	0.0369
sleep_time_hrs	focus_score	-0.179	0.1168
sleep_time_hrs	productivity_score	-0.174	0.1272
screen_time_hrs	productivity_score	0.135	0.2392
social_media_time_hrs	focus_score	-0.128	0.2646
screen_time_hrs	focus_score	-0.117	0.3058
AI_usage_encoded	focus_score	-0.115	0.3170
social_media_time_hrs	productivity_score	-0.034	0.7709
AI_usage_encoded	productivity_score	0.011	0.9215

Focus and productivity move together, as expected ($\rho = 0.486$, $p < 0.0001$). The clearest digital-habit result in the study is about the phone: **students who check their phone more during study report lower focus** ($\rho = -0.374$, $p = 0.0007$) **and lower productivity** ($\rho = -0.237$, $p = 0.0369$). Total screen time, social media time, sleep duration and AI-use frequency show no significant link to either outcome (all $p > 0.10$).

Key finding: In this sample, the *amount* of screen time shows no significant link to focus or productivity, but checking the phone more often *during study* goes with lower focus and productivity. This is in line with earlier research showing that just having a phone nearby reduces available thinking capacity (Ward et al., 2017). The data are observational, so this is an association, not proof of cause.

4.2.4 How the Habits Relate to Each Other

Apart from focus and productivity, the strongest links are between screen habits themselves: social media with total screen time ($\rho = 0.564$, $p < 0.0001$), gaming with social media ($\rho = 0.449$, $p < 0.0001$), and gaming with screen time ($\rho = 0.404$, $p = 0.0002$). Heavy digital users tend to be heavy across several habits at once, rather than swapping one for another. Part of this overlap may come from the inconsistent screen-time answers described in Section 3.3.

The matrix also shows that students who use their phone more while studying study less ($\rho = -0.287$, $p = 0.0110$), that online shopping goes with lower focus ($\rho = -0.313$, $p = 0.0052$), and that satisfaction with balance is higher for more focused students ($\rho = 0.318$, $p = 0.0046$) and lower for heavier phone users ($\rho = -0.278$, $p = 0.0137$). These p-values are for the ten strongest of the 91 pairs in the matrix and are not corrected for multiple comparisons, so the weaker ones should be treated as leads rather than confirmed findings.

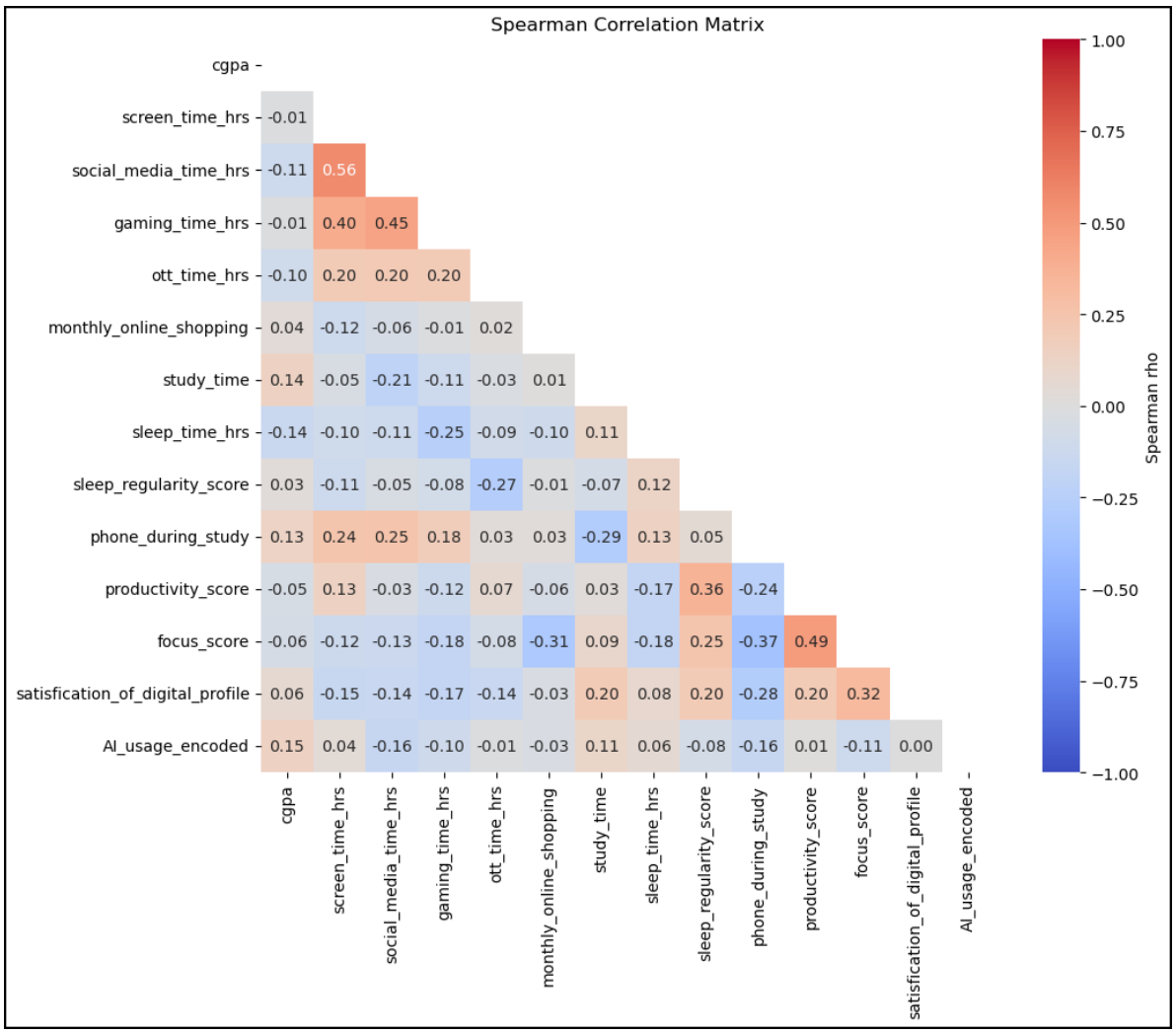


Figure 6. *Spearman correlation matrix for the numeric and rating variables (lower triangle). Red means the two variables move together (positive ρ); blue means one goes up as the other goes down (negative ρ).*

4.2.5 Time Use and AI Purposes

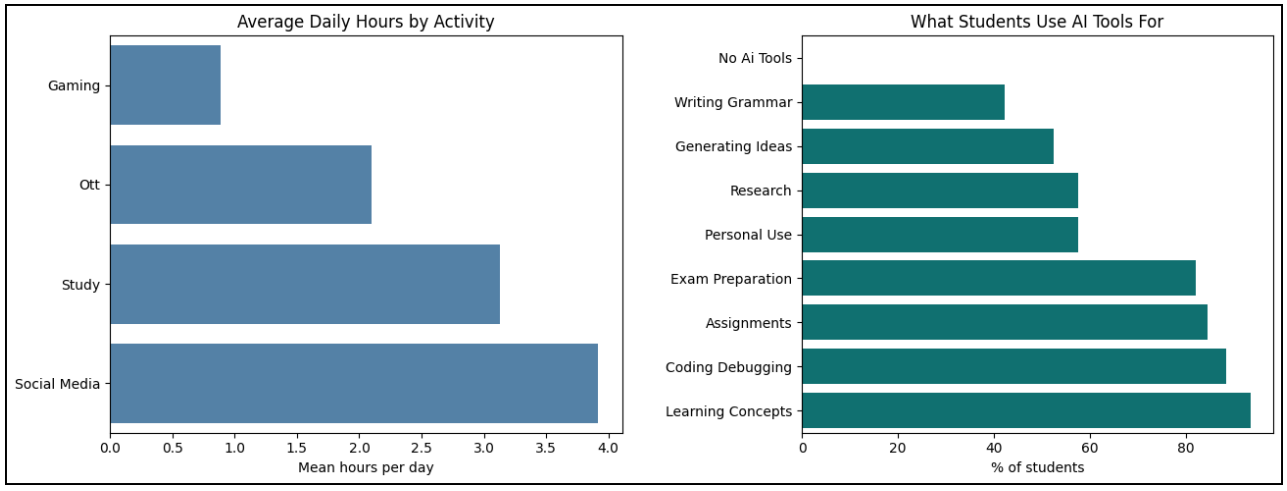


Figure 7. *Average hours per day by activity (left) and the share of students using AI tools for each purpose (right).*

On an average day, students spend more time on social media (3.92 h) than studying (3.13 h). The top AI uses are learning concepts (94%), coding/debugging (88%) and assignments (85%), then exam preparation (82%), well ahead of personal use (58%). In this sample, AI tools are mostly a study aid, not entertainment.

4.2.6 Sleep, Study Habits and Satisfaction

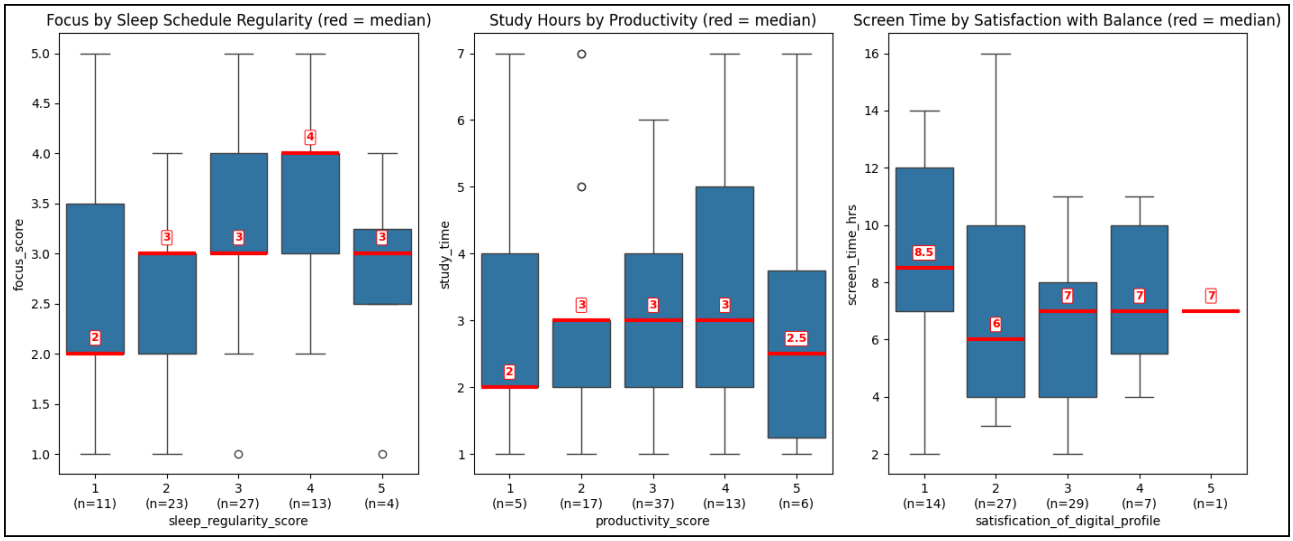


Figure 8. *Focus by sleep regularity, study hours by productivity, and screen time by satisfaction with balance. Red lines mark the median. For levels 1 to 5, the median focus is 2, 3, 3, 4 and 3; median study hours are 2, 3, 3, 3 and 2.5; median screen time is 8.5, 6, 7, 7 and 7 hours.*

- **Regular sleep goes with better focus and productivity.** A more regular sleep schedule is associated with higher productivity (ρ = 0.360, p = 0.0012) and better focus (ρ = 0.246, p = 0.0302). Sleep *length* shows no significant link to either (Table 11); only regularity was significantly associated with them, which is consistent with Phillips et al. (2017).
- **Study hours and productivity.** Students who study more are not more productive (ρ = 0.026, p = 0.8244).
- **Satisfaction with balance.** Higher screen time (ρ = −0.153, p = 0.1811) and social media time (ρ = −0.142, p = 0.2139) go with lower satisfaction, and more study time with higher satisfaction (ρ = 0.200, p = 0.0788) — none of these three is significant at 5%.

5. Conclusion

5.1 Summary of Findings

- **CGPA is not significantly linked to any single habit we tested** — screen time, social media, gaming, OTT, sleep or study time (all p > 0.05). CGPA also looks similar across AI-use groups.
- **Phone use during study is the clearest digital finding.** More phone use goes with lower focus (ρ = −0.374, p = 0.0007) and lower productivity (ρ = −0.237, p = 0.0369).
- **Focus and productivity go together** (ρ = 0.486, p < 0.0001).
- **Sleep regularity, not sleep length,** goes with productivity (ρ = 0.360, p = 0.0012) and focus (ρ = 0.246, p = 0.0302).
- **Screen habits cluster:** social media with screen time (ρ = 0.564) and with gaming (ρ = 0.449) — heavy users are heavy across several habits at once.
- **AI tools** are used by almost everyone (91% daily or more), mostly for study, and show no clear link to CGPA.
- **Balance is the weak spot:** satisfaction averages 2.41 out of 5, the lowest rating in the survey. It is higher for more focused students (ρ = 0.318, p = 0.0046) and lower for heavier phone users (ρ = −0.278, p = 0.0137); these p-values are not corrected for multiple comparisons.

5.2 Limitations

- **Small sample.** 78 responses limits statistical power.
- **Convenience sample.** Students were not randomly chosen, so results may not apply beyond our own circle (Section 2.3).
- **Inconsistent screen-time answers.** 30 of 78 responses (38.5%) report more time on social media, gaming and OTT than in total (Section 3.3). We kept them, so the screen-time results carry this uncertainty.
- **Self-reported data.** Hours, CGPA and ratings can be misremembered or rounded to look better.
- **Links, not causes.** For example, sleep regularity and productivity might both come from a third factor, such as how organised a student is.
- **Many tests, no correction.** We ran many correlations without a multiple-comparison correction, so some significant results could be chance. Treat them as leads for a bigger study.

5.3 Final Remarks

In one sentence: Total screen time and social media use showed little direct link to grades; what lines up with focus and productivity is **checking the phone less while studying** and **keeping a steady sleep schedule**. These are associations only; a larger, repeated study is needed to test whether these links are cause and effect.

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Appendix A — Questionnaire

Original questionnaire form: https://bit-n-build-bumble-bees.vercel.app/participant/experiments/9e188efd-e551-467b-84ba-d4d4264954c4/run

Table A1. The survey questionnaire.

No.	Question	Answer options
Section 1: Student Profile		
1	What is your age?	16 or below · 17 · 18 · 19 · 20 · 21 · 22 · 23 or above
2	What is your current year of study?	1st Year · 2nd Year · 3rd Year · 4th Year
3	What is your current CGPA? (10-point scale; conversion formula supplied for percentage and 5-point scales)	Slider, 0 to 10
Section 2: Digital Habits and Screen Time		
4	On a typical day, how many hours of total screen time do you have?	Slider, 0 to 16 hours
5	Approximately how many hours do you spend on social media each day?	Slider, 0 to 16 hours
6	Approximately how many hours do you spend gaming each day?	Slider, 0 to 16 hours
7	How many hours do you spend watching YouTube/OTT for entertainment each day?	Slider, 0 to 16 hours
8	How many times do you shop online in a typical month?	Slider, 0 to 30 times
9	Which activity takes up most of your free time?	Social media · Gaming · YouTube / OTT · Studying · AI tools · Online shopping · Friends / Social Activities
Section 3: AI Tool Usage		
10	How often do you use AI tools such as ChatGPT, Gemini, Claude, Copilot, etc.?	Never · Rarely · Daily · Several times a day
11	What do you mainly use AI tools for? (multi-select)	Learning concepts · Coding / debugging · Assignments · Research · Writing / grammar · Generating ideas · Exam preparation · Personal use · I don't use AI tools
Section 4: Study Habits and Sleep Patterns		
12	On average, how many hours do you spend studying outside of classes?	Slider, 1 to 16 hours
13	Approximately how many times do you check your phone during a normal study session?	Rating, 1 = Not at all to 5 = Extremely
14	How productive do you usually feel during your study time?	Rating, 1 = Very productive to 5 = Not productive
15	How focused are you usually while studying?	Rating, 1 = Very focused to 5 = Not focused
16	How many hours do you usually sleep per night?	Slider, 0 to 16 hours
17	How would you describe your usual sleep schedule?	Rating, 1 = Very regular to 5 = Very irregular
Section 5: Overall Perception		
18	How satisfied are you with your current balance between digital activities, studies and personal time?	Rating, 1 = Very dissatisfied to 5 = Very satisfied